

SUDEEP K S

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PROFESSIONAL SUMMARY

MS Applied AI candidate at Northeastern University (GPA: 3.89) with 3+ years of industry experience building AI automation systems at Capgemini. Designed and deployed a production-grade 5-agent LangGraph clinical AI platform integrating computer vision, NLP, RAG, and LLMs. Seeking AI/ML Engineer or Data Science co-op (Jan 2027) to build impactful, scalable AI systems.

TECHNICAL SKILLS

Python • PyTorch • TensorFlow • Scikit-learn • Deep Learning • Computer Vision • NLP • LLMs • RAG • LangChain • LangGraph • Hugging Face • OpenCV • FastAPI • Streamlit • Docker • MLflow • SQL • Azure • Git • Tableau

EDUCATION

Northeastern University — MPS, Applied AI | GPA: 3.89/4.0

Boston, MA

BMS Institute of Technology & Management — B.E. | CGPA: 7.2/10

2022

IIIT Bangalore — Advanced Certificate, Data Science (Deep Learning) | CGPA: 3.89/4.0

Sep 2025

PROFESSIONAL EXPERIENCE

Capgemini | Senior Software Engineer — AI Automation & RPA

Aug 2023 – Dec 2025

- Engineered RPA automation QA framework for healthcare AI workflows using Automation Anywhere 360, reducing production defects by 30% through systematic edge-case coverage and data-driven testing.
- Analyzed healthcare data pipelines and AI-driven BOT execution patterns using SQL, validating data integrity across 10+ automated clinical workflows and ensuring reliable downstream analytics.
- Designed and executed CI/CD workflows using Azure DevOps across 5 agile sprint teams, consistently delivering project milestones on time and reducing deployment errors by 25%.
- Collaborated with cross-functional teams to assess AI model impact on clinical automation systems, enabling proactive risk mitigation across 3 production healthcare platforms.

Rinex Technologies | Front-End Developer

Sep 2022 – Jul 2023

- Built 8+ responsive web applications using HTML, CSS, JavaScript, and Bootstrap, improving page load performance by 40% through code optimization and lazy loading.
- Collaborated with UX and back-end teams to implement data-driven interfaces, reducing design-to-deployment cycle time by 20%.

PROJECTS

HealthGuard AI — Multimodal Clinical Intelligence Platform — *LangGraph / LLMs / Computer Vision / RAG*

github.com/kssudeep/healthguard-ai

- Architected a 5-agent LangGraph system (Vision, NLP, RAG, Critic, Synthesizer) with supervisor-reflection pattern and circuit breaker, achieving end-to-end clinical report generation in ~70 seconds.
- Fine-tuned DenseNet-121 on NIH ChestX-ray14 (14 pathologies) with Grad-CAM explainability; implemented BioBERT NER for clinical entity extraction with negation detection using SpaCy.
- Built hybrid FAISS + BM25 retrieval pipeline with cross-encoder reranking over PubMed + clinical guidelines, improving retrieval relevance by 31% over single-method baseline.
- Deployed FastAPI backend + Streamlit UI on Docker with MLflow experiment tracking (26 metrics/run), Prometheus monitoring, and GitHub Actions CI/CD targeting Azure Container Apps.

Pilot Eye Gaze & Hand Gesture Recognition — *Computer Vision / Deep Learning*

github.com/kssudeep/Pilot-Eye-Gaze-and-hand-Gesture_Final

- Built real-time multimodal AI system combining eye gaze tracking and hand gesture recognition using OpenCV and MediaPipe, achieving <50ms inference latency at 30fps.
- Deployed contactless control interface applicable to aviation, AR/VR, and accessibility — processing dual sensor streams with deep learning-based feature fusion.

OmegaTK RAG Coding Assistant — *Commissioned by OpenEye / Cadence Molecular Sciences | LangChain / FAISS / RAG /*

Gemini

github.com/kssudeep/omegatk-rag

- Commissioned by OpenEye (Cadence Molecular Sciences) to build a RAG coding assistant for their OmegaTK Python toolkit; combined FAISS vector search with Gemini 2.5 Flash to generate executable molecular conformer code from natural language prompts.
- Engineered history-aware two-step RAG chain (condense + MMR retrieval, k=8) over 8 live OmegaTK documentation pages; delivered Streamlit chat UI with conversation history, source citations, and FAISS index rebuild controls.
- Achieved 15/15 on automated test suite validating code generation, clarification, multi-turn context, and scope enforcement; maintained structured 15-day git commit history per client SDLC requirements.